

KTG & Radiation: Gas Laws

- **Boyle's law:** If m and T are constant, $V \propto \frac{1}{P}$, which gives $P_1V_1 = P_2V_2$.

- **Charles's law:** If m and P are constant, $V \propto T$, which gives

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

- **Gay-Lussac's law:** If m and V are constant, $P \propto T$, which gives

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

- **Avogadro's law:** If P , V , and T are the same, then $N_1 = N_2$, where N_1 and N_2 are the number of molecules. Also, $V \propto N$, meaning

$$\frac{V_1}{N_1} = \frac{V_2}{N_2}$$

- **Graham's law:** If P and T are constant, the rate of diffusion $r \propto \frac{1}{\sqrt{\rho}}$, resulting

$$\text{in } \frac{r_1}{r_2} = \sqrt{\frac{\rho_2}{\rho_1}}$$

(where ρ is density).

- **Dalton's law:** Total pressure is

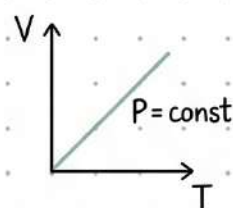
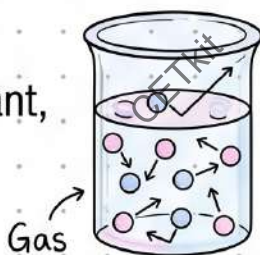
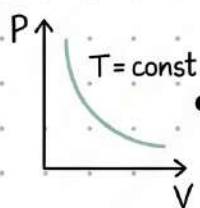
$$P = P_1 + P_2 + P_3 \dots,$$

where $P_1, P_2, P_3 \dots$ are pressures exerted by each component in the mixture.

- **Ideal Gas Equation:**

$$PV = nRT = Nk_B T$$

- n = number of moles.
- N = number of molecules.
- R = universal gas constant.
- k_B = Boltzmann's constant.

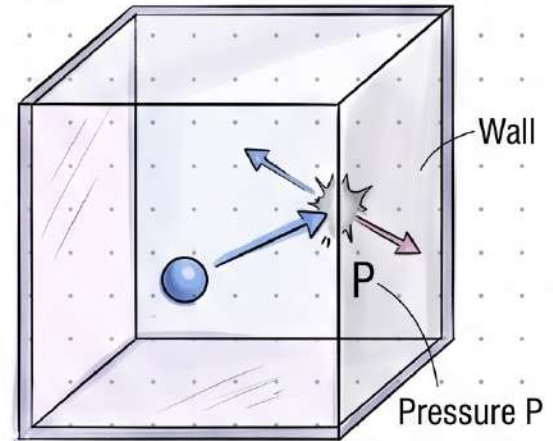


Kinetic Theory & Molecular Velocities

Pressure Exerted by Ideal Gas

$P = \frac{1}{3} \frac{mN}{V} v^2$ (where v^2 is mean square velocity and m is mass of each molecule).

Alternatively, $P = \frac{1}{3} n m v^2$, where number density $n = \frac{N}{V}$.

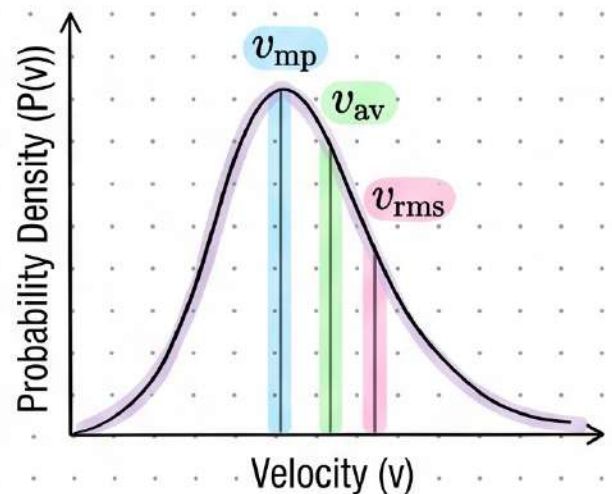


Molecular Velocities

RMS Velocity: $v_{\text{rms}} = \sqrt{\frac{3k_B T}{m}}$

Average Velocity: $v_{\text{av}} = \sqrt{\frac{8k_B T}{\pi m}}$

Most Probable Velocity: $v_{\text{mp}} = \sqrt{\frac{2k_B T}{m}}$



Kinetic Energy: $K.E. = \frac{3}{2} k_B T = \frac{1}{2} m v^2$.

Mean Free Path (ℓ): $\ell = \frac{1}{\sqrt{2} n \pi d^2}$, where d is the diameter of the molecule and n is number density.

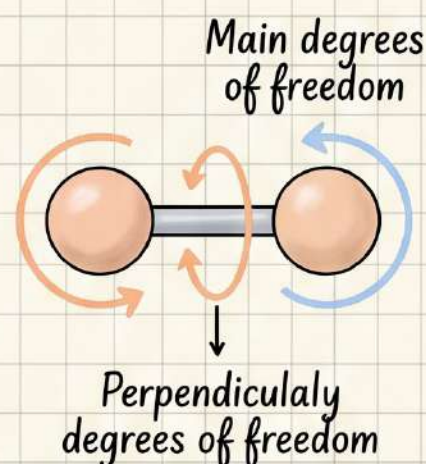
Specific Heats & Degrees of Freedom

Key Relations

$$\gamma = \frac{C_p}{C_v} \quad (\text{where } C_p \text{ is specific heat at constant pressure and } C_v \text{ is specific heat at constant volume})$$

$$C_p - C_v = R \quad (\text{where } R \text{ is the universal gas constant})$$

Gas Type	Degrees of Freedom	C_v	C_p	γ
Monoatomic	3	$\frac{3}{2}R$	$\frac{5}{2}R$	$\frac{5}{3}$
Diatomic	5	$\frac{5}{2}R$	$\frac{7}{2}R$	$\frac{7}{5}$



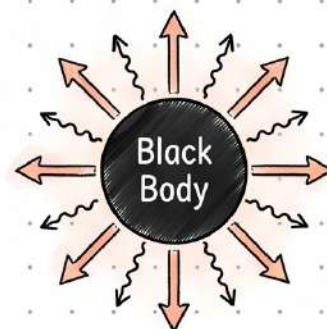
$$C_{v(\text{mix})} = \frac{n_1 C_{v1} + n_2 C_{v2}}{n_1 + n_2}$$

where n_1 and n_2 are number of moles of gases mixed together

Thermal Radiation & Laws of Cooling

→ Radiation Properties:

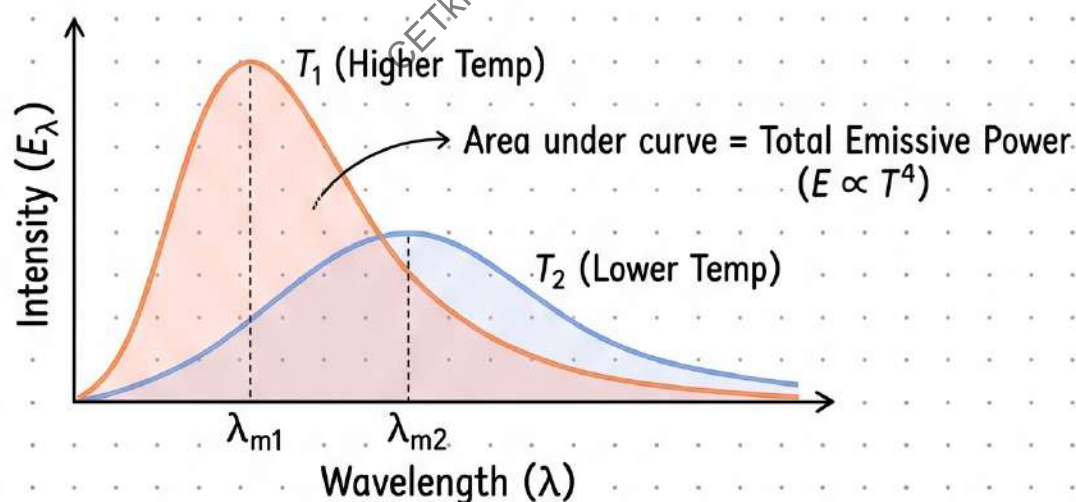
- Absorptive Power (a): $a = \frac{\text{Energy Absorbed}}{\text{Energy incident}}$
- Emissive Power (E): Energy radiated per unit area, per unit time, and per unit solid angle: $E = \frac{\Delta U}{\Delta A \cdot \Delta \omega \cdot \Delta t}$
- Spectral Emissive Power (E_λ): $E_\lambda = \frac{dE}{d\lambda}$
- Emissivity (e): $e = \frac{E}{E_0}$, where E_0 is the emissive power of a black body.



→ Stefan-Boltzmann's Law:

Radiation: $\frac{dQ}{dt} = e\sigma AT^4$ (where T is body's temperature)

Absorption: $\frac{dQ}{dt} = e\sigma AT_0^4$ (where T_0 is surrounding temperature)



Wien's Displacement Law: $\lambda_m \cdot T = b$, where $b = 0.282 \text{ cm-K}$

→ Laws of Cooling:

Net Radiation: $\frac{dQ}{dt} = e\sigma A(T^4 - T_0^4)$

Newton's Law of Cooling: $\frac{dT}{dt} \propto \Delta T$

Approximation: $\frac{T_1 - T_2}{t} = \alpha \left[\frac{T_1 + T_2}{2} - T_0 \right]$, where T_1 is initial temperature, T_2 is final temperature, and α is a constant.